



L7b: Utility-Based Portfolio Allocation

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Objectives for Today

By the end of this lecture, you should be able to:

- **Describe preferences with utility functions:** explain how a utility function ranks alternatives and how a concave utility of wealth describes risk aversion. Derive the risk-free fraction that maximizes mean-variance utility, and infer a client's risk aversion.
- **Solve budget-constrained utility problems:** solve the Cobb–Douglas problem over share counts in closed form, and state how the CES elasticity moves the allocation.
- **Compute preference weights from the single index model:** combine intercepts and betas with two market inputs from SPY. Signs choose the basket, and magnitudes split the budget.

Lectures and Examples

Lecture: Utility-Based Portfolio Allocation

Example 1: Choose the Optimal Point on the Capital Allocation Line

Which risk-free fraction maximizes utility? Infer the risk aversion behind each interview answer.

Example 2: Build a Maximum-Utility Portfolio Allocator

Which firms belong in the basket today, and how much of each?

Example 1 runs on the firms and risk-free fraction from the L6b client interview, or on defaults when the interview files are absent.

Concept Review: The Capital Allocation Line

A **complete portfolio** c holds w_f in the risk-free asset and $1 - w_f$ in the tangent portfolio $\mathcal{T}$:

$$\mathbb{E}[g_c] = g_f + (1 - w_f)(\mathbb{E}[g_{\mathcal{T}}] - g_f) \quad \text{expectation is linear}$$

$$\sigma_{g,c}^2 = (1 - w_f)^2 \sigma_{g,\mathcal{T}}^2 \quad \text{the risk-free asset adds no variance}$$

Lending ($0 < w_f < 1$) and borrowing ($w_f < 0$) move c along the capital allocation line, whose slope is the Sharpe ratio of $\mathcal{T}$.

By **two-fund separation**, investors with the same inputs who can lend and borrow freely at g_f hold the same risky fund $\mathcal{T}$. Their risk preferences set only w_f , and the L6b client interview asked for w_f directly.

Which w_f should a given investor choose?

Utility Functions and Rational Choice

Utility functions describe a decision-maker's preferences for different choices, outcomes, or states of the world. Utility is a proxy for satisfaction, so a larger utility is always better.

Utility function: an agent chooses from alternatives $\mathcal{X} = \{x_1, \dots, x_n\}$, e.g., bundles of goods, portfolios, or states of the world. $U : \mathcal{X} \rightarrow \mathbb{R}$ assigns each alternative a number, in *utils*, that ranks the agent's preferences:

$$U(x_i) > U(x_j) \iff x_i \succ x_j \quad \text{and} \quad U(x_i) = U(x_j) \iff x_i \sim x_j$$

- **Ordinal:** utility ranks alternatives but does not measure how much better one is. Any order-preserving transformation of U , e.g., $\ln U$ for positive U , describes the same preferences.
- **Key idea:** when faced with competing alternatives, a rational decision-maker always chooses the alternative with the largest utility.

Common Utility Functions

Utility functions, and their parameters, are subjective. Five common forms for a bundle of goods $\mathbf{x} = (x_1, \dots, x_n)$ with $x_i \geq 0$ and $\alpha_i > 0$:

Utility function	Form	Models
Linear	$U(\mathbf{x}) = \sum_i \alpha_i x_i$	goods that are perfect substitutes
Logarithmic ($\beta > 0$)	$U(\mathbf{x}) = \ln(\sum_i \alpha_i x_i + \beta)$	diminishing satisfaction from more goods
Cobb–Douglas	$U(\mathbf{x}) = \prod_i x_i^{\alpha_i}$	goods held together
Leontief	$U(\mathbf{x}) = \min(x_1/\alpha_1, \dots, x_n/\alpha_n)$	perfect complements, fixed proportions
CES ($\eta > 0, \eta \neq 1$)	$U(\mathbf{x}) = \left(\sum_i \alpha_i x_i^{(\eta-1)/\eta}\right)^{\eta/(\eta-1)}$	substitution set by the elasticity η

We use the Cobb–Douglas and CES utilities later today. As η changes, CES moves between the linear ($\eta \rightarrow \infty$), Cobb–Douglas ($\eta \rightarrow 1$, with $\sum_i \alpha_i = 1$), and Leontief ($\eta \rightarrow 0$) forms.

Risk and Risk Aversion

When outcomes are uncertain, a rational decision-maker ranks choices by the **expected utility** $\mathbb{E}[U(W)]$ (**von Neumann–Morgenstern**). The shape of the utility of wealth $U(W)$ sets the attitude toward risk:

Curvature of U	Risky versus certain	Risk attitude
Concave ($U'' < 0$)	$\mathbb{E}[U(W)] \leq U(\mathbb{E}(W))$	risk averse
Linear ($U'' = 0$)	$\mathbb{E}[U(W)] = U(\mathbb{E}(W))$	risk neutral
Convex ($U'' > 0$)	$\mathbb{E}[U(W)] \geq U(\mathbb{E}(W))$	risk seeking

The **certainty equivalent** is the guaranteed wealth with the same utility, $U(\text{CE}) = \mathbb{E}[U(W)]$. The **risk premium** is the expected wealth a decision-maker gives up to remove the risk:

$$\text{RP} = \mathbb{E}(W) - \text{CE}$$

For a risk-averse decision-maker, $\text{CE} \leq \mathbb{E}(W)$ and $\text{RP} \geq 0$.

Arrow–Pratt Measures

How risk averse is a decision-maker? For a twice-differentiable utility of wealth with $U'(w) > 0$, Arrow and Pratt (1964) define the coefficients of absolute and relative risk aversion:

$$r(w) = -\frac{U''(w)}{U'(w)} \quad \text{and} \quad \bar{r}(w) = w r(w)$$

	$U(w)$	$r(w)$	$\bar{r}(w)$
Log	$\ln w$	$1/w$	1
Linear	$a w + b$	0	0
Power	w^τ	$(1 - \tau)/w$	$1 - \tau$

- Risk averse: $r(w) > 0$. Risk neutral: $r(w) = 0$. Risk seeking: $r(w) < 0$.
- Log and power utilities have constant relative risk aversion.

Why it matters for a portfolio: for a small risk, $RP \approx \frac{1}{2} r(\mathbb{E}(W)) \text{Var}(W)$. Per unit of wealth, and for small returns, the certainty-equivalent return is about the expected return minus $\bar{r}/2$ times the return variance. Mean-variance utility has this form, with A in the role of $\bar{r}$.

Mean-Variance Utility

For a complete portfolio, mean-variance utility rewards expected growth and charges for risk:

$$U(w_f) = \underbrace{\mathbb{E}[g_c]}_{\text{reward}} - \underbrace{\frac{A}{2} \Delta t \sigma_{g,c}^2}_{\text{risk penalty}}$$

$A > 0$ is the risk-aversion coefficient (dimensionless). $\Delta t \sigma_{g,c}^2$ is the squared volatility (L5a), so both terms are in year^{-1} . Along the line, both terms depend only on w_f :

$$U(w_f) = g_f + (1 - w_f)(\mathbb{E}[g_{\mathcal{T}}] - g_f) - \frac{A}{2}(1 - w_f)^2 \Delta t \sigma_{g,\mathcal{T}}^2 \quad \text{substitute the line}$$

$$\frac{dU}{dw_f} = -(\mathbb{E}[g_{\mathcal{T}}] - g_f) + A(1 - w_f) \Delta t \sigma_{g,\mathcal{T}}^2 \quad \text{differentiate}$$

$$0 = -(\mathbb{E}[g_{\mathcal{T}}] - g_f) + A(1 - w_f^*) \Delta t \sigma_{g,\mathcal{T}}^2 \quad \text{first-order condition}$$

The Optimal Complete Portfolio

Solving the first-order condition for the risky fraction gives:

$$1 - w_f^* = \frac{\mathbb{E}[g_{\mathcal{T}}] - g_f}{A \Delta t \sigma_{g, \mathcal{T}}^2}$$

- **A maximum**: the second derivative, $-A \Delta t \sigma_{g, \mathcal{T}}^2$, is negative. Doubling A halves the risky fraction.
- **Lend or borrow**: the investor with $A_{\mathcal{T}} = (\mathbb{E}[g_{\mathcal{T}}] - g_f)/(\Delta t \sigma_{g, \mathcal{T}}^2)$ holds $\mathcal{T}$ alone. $A > A_{\mathcal{T}}$ lends, and $A < A_{\mathcal{T}}$ borrows.

Indifference Curves and Risk Aversion

Where does the optimum sit on the line? An indifference curve joins the portfolios with the same utility. Holding U fixed gives a parabola that opens upward in the risk-return plane:

$$\mathbb{E}[g_c] = U + \frac{A}{2} \Delta t \sigma_{g,c}^2$$

Higher curves mean higher utility, and the optimal complete portfolio is where the highest curve the investor can reach touches the line.

The same result runs backward. A client who picks a fraction w_f reveals the coefficient that makes it optimal:

$$A = \frac{\mathbb{E}[g_{\mathcal{T}}] - g_f}{(1 - w_f) \Delta t \sigma_{g,\mathcal{T}}^2}$$

Example 1: Choose the Optimal Point on the Capital Allocation Line

Utility over Holdings

Every investor so far holds the same risky fund. Instead, ask a consumer-choice question: given a budget and today's prices, which holdings does the investor prefer most? At time t , with prices $S_i(t)$ (USD per share) and wealth $W_{\mathcal{P}}(t)$ (USD), choose the share counts $n_i \geq 0$:

$$\underset{n_1, \dots, n_{|\mathcal{P}|}}{\text{maximize}} \quad U(n_1, \dots, n_{|\mathcal{P}|}) \quad \text{subject to} \quad \sum_{i \in \mathcal{P}} n_i S_i(t) = W_{\mathcal{P}}(t)$$

This is a **holding utility**, a preference over the shares held today at today's prices. It has no variance term. Risk enters through the preference weights, which depend on the SIM betas and the state of the market.

Cobb–Douglas Utility

The exponents are the **preference weights** $\gamma_i \in (-1, 1)$. The investor maximizes the product over every asset, subject to the budget and a small floor of $n_{\min} > 0$ shares on each holding:

$$\begin{array}{ll} \underset{n_1, \dots, n_{|\mathcal{P}|}}{\text{maximize}} & U(n_1, \dots, n_{|\mathcal{P}|}) = \prod_{i \in \mathcal{P}} n_i^{\gamma_i} \\ \text{subject to} & \sum_{i \in \mathcal{P}} n_i S_i(t) = W_{\mathcal{P}}(t) \quad (\text{budget constraint}) \\ & n_i \geq n_{\min} \quad \forall i \in \mathcal{P} \quad (\text{share floor}) \end{array}$$

The floor keeps the utility finite, because for $\gamma_i < 0$ the term $n_i^{\gamma_i}$ grows without bound as $n_i \rightarrow 0$. The sign of γ_i splits the assets in two: the **preferred set** $\mathcal{A}^+ = \{i : \gamma_i > 0\}$ is the basket, and the rest form the **non-preferred set** $\mathcal{A}^- = \{i : \gamma_i \leq 0\}$.

Cobb–Douglas Utility: Floors and Net Budget

Only the ranking matters, so $\ln U$ has the same maximizer as U . We set the floors first:

$$\frac{\partial \ln U}{\partial n_k} = \frac{\gamma_k}{n_k} \leq 0 \quad k \in \mathcal{A}^-, \text{ so more shares never raise } \ln U$$
$$n_k^* = n_{\min} \quad \text{spend every dollar above the floor on } \mathcal{A}^+$$

Substituting the floors into the budget constraint leaves the **net budget** for the preferred set:

$$\sum_{i \in \mathcal{A}^+} n_i S_i(t) + n_{\min} \sum_{k \in \mathcal{A}^-} S_k(t) = W_{\mathcal{P}}(t)$$
$$\sum_{i \in \mathcal{A}^+} n_i S_i(t) = W_{\mathcal{P}}(t) - n_{\min} \sum_{k \in \mathcal{A}^-} S_k(t) \equiv W_{\text{adj}} > 0$$

A Lagrange multiplier λ then splits W_{adj} over the preferred set, assuming each preferred share count stays above its floor.

Cobb–Douglas Utility: Allocation

$$\mathcal{L} = \sum_{i \in \mathcal{A}^+} \gamma_i \ln n_i - \lambda (\sum_{i \in \mathcal{A}^+} n_i S_i(t) - W_{\text{adj}})$$

$$0 = \gamma_i / n_i - \lambda S_i(t)$$

$$n_i S_i(t) = \gamma_i / \lambda$$

$$W_{\text{adj}} = \sum_{j \in \mathcal{A}^+} \gamma_j / \lambda$$

$$1/\lambda = W_{\text{adj}} / \sum_{j \in \mathcal{A}^+} \gamma_j$$

floor terms are constant

differentiate with respect to n_i

dollars in proportion to γ_i

sum over the preferred set

solve for the multiplier

Substituting $1/\lambda$ into $n_i S_i(t) = \gamma_i / \lambda$ gives:

$$n_i^* = \underbrace{\frac{\gamma_i}{\sum_{j \in \mathcal{A}^+} \gamma_j}}_{\text{preference share}} \cdot \underbrace{\frac{W_{\text{adj}}}{S_i(t)}}_{\text{budget in shares}} \quad (i \in \mathcal{A}^+), \quad n_i^* = n_{\min} \quad (i \in \mathcal{A}^-)$$

If a preferred count falls below $n_{\min}$, pin it at the floor and solve again with the rest.

The dollar weight within W_{adj} is $\gamma_i / \sum_j \gamma_j$.

CES Utility: A Concentration Dial

The elasticity $\eta > 0$ controls how willingly the allocator trades one preferred asset for another. For $\eta \neq 1$:

$$U_{\text{CES}} = \left(\sum_{i \in \mathcal{A}^+} \gamma_i n_i^{(\eta-1)/\eta} \right)^{\eta/(\eta-1)}$$

$$n_i^* = W_{\text{adj}} \frac{(\gamma_i/S_i(t))^\eta}{\sum_{j \in \mathcal{A}^+} S_j(t)(\gamma_j/S_j(t))^\eta} \quad (i \in \mathcal{A}^+)$$

Share counts are proportional to the *bang for the buck* $\gamma_i/S_i(t)$ raised to η , with the same floor check. Three limits:

- **Unit elasticity** ($\eta \rightarrow 1$): the Cobb–Douglas allocation.
- **Large elasticity** ($\eta \rightarrow \infty$): every other asset falls to its floor, and the rest goes to the largest $\gamma_i/S_i(t)$, when unique.
- **Small elasticity** ($\eta \rightarrow 0^+$): equal share counts of every preferred asset.

Proofs: [CES companion](#), which also shows that CES dollar weights depend on share price levels when $\eta \neq 1$.

Preference Weights from the Single Index Model

Each trading day, from the L6b SIM parameters (α_i, β_i) , the recent market growth rate $\tilde{g}_{M,t}$, and the market-state signal ξ_t :

$$\gamma_i(t) = \tanh \left(\frac{\alpha_i}{\beta_i^{\xi_t}} + \beta_i^{1-\xi_t} \tilde{g}_{M,t} \right)$$

The $\tanh$ keeps the weight in $(-1, 1)$. The argument equals $(\alpha_i + \beta_i \tilde{g}_{M,t}) / \beta_i^{\xi_t}$, the SIM's expected growth of asset i in today's market divided by $\beta_i^{\xi_t}$. The model assumes $\beta_i > 0$.

- **Which assets belong in the basket?** $\gamma_i(t) > 0$ exactly when $\tilde{g}_{M,t} > -\alpha_i / \beta_i$. A weak market keeps only the assets with the lowest thresholds. If every $\gamma_i(t) \leq 0$, hold the floors plus cash.
- **How much of each?** The magnitude sets the budget share. A bearish signal ($\xi_t > 0$) tilts toward low-beta names, and a bullish one toward high-beta names.

The Market Inputs

Both come from SPY through the EMA of L5a, $\bar{S}_t = \omega S_t + (1 - \omega)\bar{S}_{t-1}$ with $\omega = 2/(L + 1)$. With $L_{\text{short}} = 21$ and $L_{\text{long}} = 63$ days and a gain $G > 0$, the signal is the scaled crossover:

$$\xi_t = -G \left(\frac{\bar{S}_t^{\text{short}}}{\bar{S}_t^{\text{long}}} - 1 \right)$$

- Falling market: $\xi_t > 0$ (bearish). Rising market: $\xi_t < 0$ (bullish). G makes $|\xi_t| \leq 1$ on 99% of training days.
- $\tilde{g}_{M,t}$: the EMA ($L_{\text{growth}} = 10$ days) of $g_{M,t} = \ln(S_t/S_{t-1})/\Delta t$. It swings by about one per year, several times the intercepts, so assets cross their thresholds.
- Both inputs on day t use prices through day $t - 1$.

Example 2: Build a Maximum-Utility Portfolio Allocator

Optional Advanced Material

The [advanced index](#) describes the optional notebook:

- [CES limits and log-linear utility](#): derive the closed-form CES optimum and prove its three limits. Show that log-linear utility has the Cobb–Douglas optimizer, and why CES weights depend on share price levels.

Summary

- **Risk aversion picks the point on the capital allocation line:** we maximized mean-variance utility along the line and found the optimal risky fraction in closed form. More risk-averse investors lend more, and a client's chosen fraction reveals a risk-aversion coefficient.
- **Cobb–Douglas turns preferences into weights in closed form:** the preferred budget is split in proportion to the positive preference weights. The CES elasticity moves the allocation between concentration and equal share counts.
- **The single index model and the market set the preferences:** the recent market growth decides which assets belong in the basket, and the crossover signal tilts the budget between high-beta and low-beta assets.

Risk aversion picks the point on the capital allocation line,
and preference weights pick the assets and their weights.